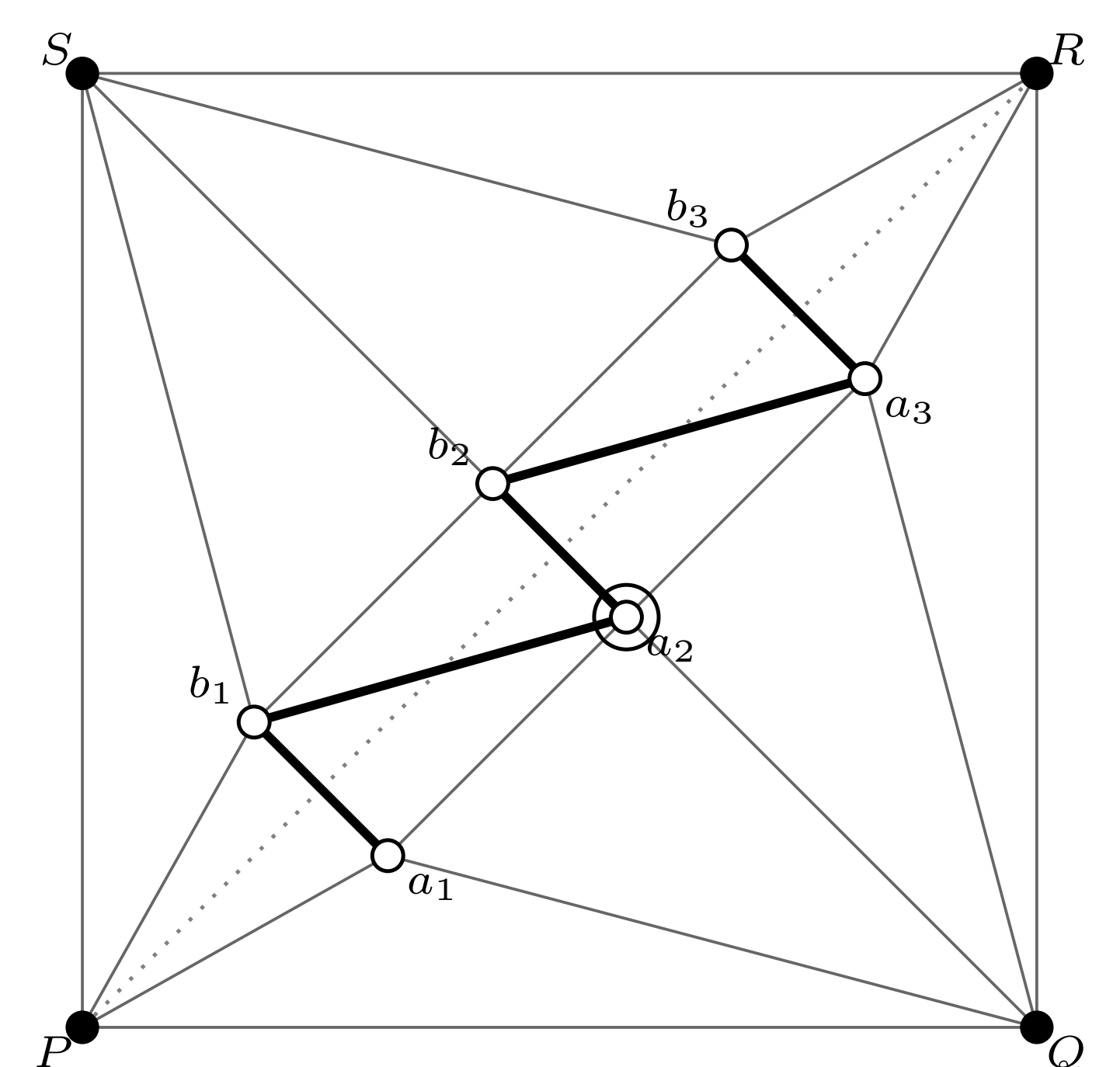

$N(a_2)=\{Q, a_1, a_3, b_1, b_2\}$